Project Carrier Course for Medical Engineers, part 1 (CM2015)

Lab 7 – Basic measurements

Introduction

One step in the CE marking and certification of medical technology is to prove that it is safe and efficient. For a medical device that measures medical signals, one sub-part is in defining how close the measurement is to the actual value. In this exercise, your task is to define that for the medical device that you built in lab 1, the heart rate measurement tool based on a PPG-sensor. Obviously, we are not going to mass produce the tool that you made, so we will skip the step regarding making sure that the results of this study hold true for all manufactured devices. Instead, we simplify and only consider the device that you created.

Note: It is **not** important that your lab 1 solution does high quality measurements in this exercise. No, here, we want to find out exactly how good (or bad) the measurements are.

Designing tests

The first part in CE-marking is to define the 'Intended Use' of the device. This is a text describing what the device is supposed to be used for and by whom. If a medical device is used in accordance with its intended use, it must fulfill a range of criteria, including efficient. E.g., report a good enough measurement result. If the device is used for something else, the liability is moved from the manufacturer to the user.

For measurements, we need to define what is the best possible measurement we can do, something often known as the gold standard. This is what we want to compare against. Either we assume that the measurement done by the gold standard is exactly correct, or it has known uncertainty limits that we know.

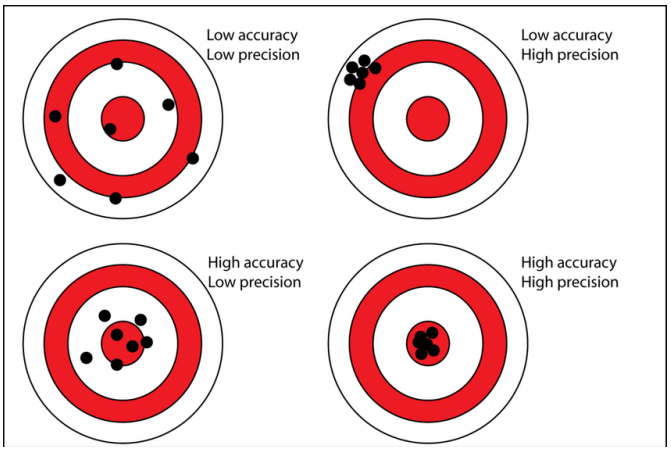
What statistical guarantees that we need to make? We need to prove that the measurement accuracy is good enough when used in accordance with the intended use. How can we define this in statistical terms?

Finally, we need to design exactly which tests we need to carry out in order to calculate the statistical terms and verify whether the device can be used for its intended use. Typically, for devices that measure something, we design a measurement campaign. We define what should be measured as a test plan. The test plan must include things, such as:

- Artificial signals or real test persons or both?
- How many measurements? Or how often?
- A single device? Several different devices? Devices from different manufacturing batches?
- For how long? Over a few minutes or over several months?
- How many test persons? Demographics of the test persons (adults, age, sex, skin color, healthy/co-morbidities, etc).
- Who is doing the measurement (the test leader, a health worker, a family member, or the patient him/herself)?
- Over what measurement range are testing? Can we measure the extremes somehow?
- Test under conditions that may influence the measurement results? E.g., different ambient temperature (room temp. or -15°C), in dark/daylight, under a blanket, etc.

Statistical tools to consider

The statistical tools we may consider is mainly accuracy (bias) and precision (low variance). Often the difference is illustrated with the figure below:



Another way to model the same thing is as follows:

$$X = T + e_s + e_r$$

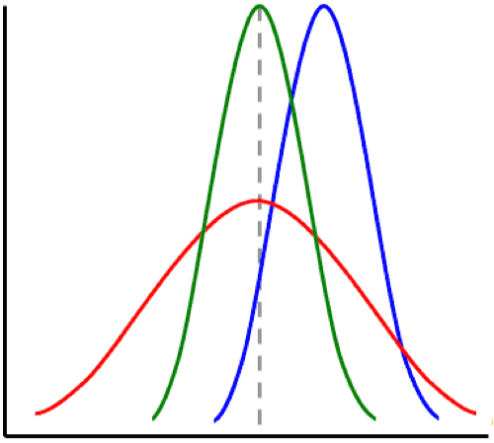
where

$e_s \in \boldsymbol{R}$ (constant)

$e_r \in N(0, \sigma)$ (random)

T is the actual measurand

Usually, a measurement can be modelled as Gaussian distributed (Normal). The figure below describes three of the four cases above if the gray dashed line is the ground truth. Namely no bias, good precision (green), no bias, poor precision (red), and some bias, good precision (blue).



We usually can compensate the bias if it is known. If the bias is different for different manufactured devices, we talk about calibrating the device. It means that we find out the bias of that particular device and configure the device to subtract the bias before presenting the measurement result. Many modern devices contain an automatic calibration procedure when they are turned on. A kitchen scale is a typical example. When you turn on the kitchen scale without any load, it finds its zero-gram point.

Obviously, e_s, which is the bias, is not always a constant, but a function of factors that may influence the measurement results. For instance, in electronics, the temperature of the analog circuits can have an influence. If we do not know how this affects the result or we do not know the circuit temperature, we have to model it as a random error, i.e., as part of e_r. If we know the temperature, we may compensate for the temperature-induced bias by assuming $e_s = f(temp) + e_{ss}$, where $f(temp)$ describes the temperature-induced bias based on the temperature measurement, e_{ss} is the remaining (unexplained) bias. The same can be done for other factors as long as we can measure them and know how they affect the measurements.

Summary statistics:

- Central representation: mean, median, mode, etc.
- Spread representation: variance, standard deviation, min/max range, quantiles

We can use percentiles to define claims, such as: “the measurement error is less than ±2 bpm with a probability of 99%”.

When calculating summary statistics based on a sample, such as the mean absolute error, or the error variance, we usually want to know the confidence of the result. It is therefore crucial to also calculate the confidence intervals. Confidence intervals are often calculated based on the assumption that the random variable is normally distributed. Remember that the central limit theorem (CLT) states that: “A sum of random independent values tends toward a normal distribution“. This means that both the mean and the variance tends to a normal distribution regardless of the distribution of the measurement values. However, for small sample sizes, we cannot assume that CLT holds. A good rule of thumb is to use a sample size of at least n=20.

The Assignment

This assignment is based on the medical device that you made in lab 1. I.e., the heart rate measurement device based on the PPG-sensor.

1. Define the intended use of your lab 1 device. Here, assume that we want to measure the heart rate of healthy adult individuals in non-mobile situations. If necessary, specify the intended use further as you like.
2. Define statistical terms that prove that the device is good enough for the intended use.
3. Design a measurement study of the accuracy (bias) and precision (variance) of your device.
4. Then carry out some parts of the measurement campaign. If testing on real persons, use yourself only.
5. Report your findings.

It is okay to report that your lab 1 solution does not fulfill its intended use. This is not a requirement to pass this assignment. If your sensor is too bad to perform any useful measurements, you can skip those measurements and only perform measurements based on artificial signals. Or borrow a sensor from another pair if possible.

In point 4 above, you need to collect at least 40 measurement readings.

Additional question (not to be answered): If someone decides to use the medical device for something else than the intended use, such as measuring the heart rate variability (HRV). Would it still be efficient? What is the quality of the HRV measurements? Can this be determined without doing additional tests?

The Report

Write a report containing all the steps above and hand-in in Canvas.

Aim for a report with 4 pages, including tables and figures.

Please include references to the online material that you used when working on this exercise. You can include anything, including book chapters, online tutorials, YouTube clips, websites, Wikipedia, etc.